

Spatial ML - class 13 - Measuring spatial agglomeration

Total points 100/100



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What is spatial agglomeration? *

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- ☐ Random distribution of points
- ☐ Uniform density across all regions
- ☒ Concentration of observations in some areas and scarcity in others
- ☐ Equal distribution of observations across space



In spatial terms, high spatial entropy indicates:

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- ☒ Uniform spatial distribution
- ☐ Strong agglomeration
- ☐ Strong clustering
- ☐ No observations

What does low entropy imply in spatial data?

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- ☐ Absence of spatial structure
- ☐ High randomness
- ☐ Equal probability of locations
- ☒ Strong concentration in specific locations



Why is Voronoi tessellation used in ETA?

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- ☐ To calculate regression coefficients
- ☒ To divide space into areas representing influence of points
- ☐ To smooth spatial data
- ☐ To convert distances into probabilities

What does relative entropy (Hrel) measure?

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- ☐ Absolute number of observations
- ☒ Difference between empirical and uniform spatial distribution
- ☐ Distance between points
- ☐ Number of clusters



When Hrel is close to 1, the distribution is: * 10/10

- ☐ Highly clustered
- ☐ Completely random
- ☒ Highly uniform
- ☐ Strongly polarized

In the SPAG index, what does the distance component represent? * 10/10

- ☐ Entropy of spatial distribution
- ☐ Overlap of firm sizes
- ☐ Number of firms in a region
- ☒ Comparison of observed vs uniform average distances



What does the overlap index in SPAG measure?

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- ☐ Distance between clusters
- ☒ Degree to which firm areas overlap instead of covering space
- ☐ Coverage of territory by firms
- ☐ Number of spatial units

How is the SPAG index constructed?

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- ☐ Sum of entropy and distance
- ☐ Ratio of overlap and entropy
- ☒ Product of distance index and overlap index
- ☐ Average of all spatial measures



What is the role of firm size in SPAG?

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- ☐ It only affects entropy
- ☐ It is ignored
- ☐ It determines the number of clusters
- ☒ It influences spatial reach and overlap effects

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